

YASH JAKHOTIYA

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Backend ML engineer at **Palantir** working across **customer-embedded** agentic AI on the frontline, Foundry **product engineering** throughout the platform, and the **model-serving infrastructure** underneath. Ex-Amazon Applied Science, D.E. Shaw. **Georgia Tech MS-CS, ML**; LLM-editing thesis.

EXPERIENCE

PALANTIR

NEW YORK

Machine Learning Engineer, Backend

July 2023 - Present

- FDE rotation at **Panasonic Energy of North America**'s Nevada Gigafactory. Shipped **Ask Atom**, an AIP copilot that mines **1M+** maintenance tickets of tribal knowledge and loops new fixes back. Cut machinery downtime **75%**; anchored a 3-year **\$60M** expansion.
- Productized identified gaps in Palantir Foundry with new **Pipeline Builder** nodes. "**Use LLM**" and "**Text-to-Embeddings**" now serve **250+ organizations**, **11B+ tokens** and **52M+ embeddings** monthly.
- Returned to the frontline with **audio-to-audio** workflows for a hospital and a telecom client, converting to another eight-figure contract and seeding an **ElevenLabs partnership**. Back on backend, sped up Foundry **audio transcription 6-7x** with diarization support.
- In the process, **bootstrapped** Palantir's new **Rust + PyO3 in-house model-inference** service responding at **34-45 ms** per query.
- Branched off to using that to **enable agentic loops** by supporting hybrid OCR, layout-aware and VLM **doc extraction**.
- Extended into **agentic image search** by introducing and threading image embeddings through Foundry's indexing and retrieval services.
- Now, **integrating live media streaming** in Foundry. Byte-level streaming media is becoming a first-class primitive supporting ingest, storage, playback and out-of-the-box transformations in low-latency pipelines.

AMAZON

PALO ALTO

Applied Scientist Intern

May 2022 - August 2022

- Worked with the **Visual Search and Augmented Reality** team on customer query-to-catalog image retrieval.
- Improved Amazon's **Shop-the-Look** recall@10 by **24%** using high-attention patches in large-scale vision **transformers**.

D.E. SHAW

HYDERABAD

Member Technical, Quant Systems

July 2020 - July 2021

Summer Intern

May 2019 - July 2019

- Developed **RBAC** and config validation for the **Elastic** stack and authored OCI images for **Kubernetes**, managing **3000+** Linux hosts.
- As an intern, automated infrastructural alert assignments using **NLP**, cutting the team's workload **86%**.

KUBEFLOW, GOOGLE CLOUD

REMOTE

Google Summer of Code Student

June 2020 - August 2020

- Deployed efficient **MLOps** workflows using all six components of Kubeflow in ML pipeline **notebooks** for two NLP tasks.

INDIAN INSTITUTE OF SCIENCE

BANGALORE

Summer Research Intern

May 2018 - July 2018

- Modeled **human dance sequences** at IISc's **Vision and AI Lab** using **RNNs** and **GANs** with **<0.1** normalized L2 reconstruction loss.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

ATLANTA

MS, Computer Science · GPA 4.0/4.0

August 2021 - May 2023

- Concentration in **Machine Learning**. Research with **Prof. Zsolt Kira** on continual learning, novel category discovery and large language model editing. MS thesis: *Scalable Large Language Model Editing*.

COLLEGE OF ENGINEERING, PUNE

B. Tech, Computer Engineering · GPA 9.2/10

August 2016 - May 2020

- Minor in **Financial Engineering**. Represented the college thrice in **ACM ICPC** regionals.

PUBLICATIONS

- *Improving Assistive Robotics with Deep Reinforcement Learning*. NeurIPS 2022 Deep RL workshop.
- *Adversarial Attacks on Transformers-Based Malware Detectors*. NeurIPS 2022 **Machine Learning Safety** workshop.
- *It Takes One to Know One? Idiomatcity Detection using Zero and One-shot Learning*. NAACL 2022 SemEval workshop.

SELECTED PROJECTS

- **COEP's 2nd Student Satellite Initiative**. Wrote a **BCH** error-correction module to counter space-radiation bit flips.
- **Mutual Information Regularization**. **Contributed** the first-known **PyTorch** implementation of the **KSG** MI estimator.
- **Open-source editor**. Extended **shnupta/bric**, an open-source code editor, with UNIX's **Exuberant Ctags**-based code navigation.

LEADERSHIP

- Founded the **Association of Students of CE and IT, COEP**. A platform for technical dialogue; organized talks, contests and tutorials mentoring younger students on competitive coding and open-source software.

TECHNICAL SKILLS

- **Languages**: Java, Rust, Python, Go, Scala, SQL, Linux shell, C/C++
- **ML / AI**: PyTorch, JAX, HuggingFace, PyO3, Transformers; model serving; RAG with chunking, embedding, reranking, query expansion, evals; agents, harnesses and tool use; prompt engineering
- **Backend & Infra**: Undertow, Jersey, Conjure, Django; Spark, Flink, Kafka, Elasticsearch; Kubernetes, Docker, AWS, GCP, Palantir Apollo, Prometheus, Grafana, Git